

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026
Homework #7 Solutions

1. (a) We need to compute the likelihoods for the observation $(X, Y, Z) = (5.1, 3.2, 1)$ and choose the class that has the larger value.

$$\begin{aligned} f_H(x) &= \frac{1}{\sqrt{2\pi(1)}} e^{-\frac{(5.1-5)^2}{2(1)}} \approx 0.397, & f_S(x) &= \frac{1}{\sqrt{2\pi(1.2)}} e^{-\frac{(5.1-4.5)^2}{2(1.2)}} \approx 0.313, \\ f_H(y) &= 0.25 e^{-0.25 \cdot 3.2} \approx 0.1123, & f_S(y) &= 0.35 e^{-0.35 \cdot 3.2} \approx 0.1142. \end{aligned}$$

Since everything is independent, the overall joint likelihoods are:

$$\begin{aligned} f_H(x, y, z) &= f_H(x) f_H(y) P(z=1|H) \\ &= 0.397 \cdot 0.1123 \cdot 0.15 \\ &\approx 0.00669 \\ f_S(x, y, z) &= f_S(x) f_S(y) P(z=1|S) \\ &= 0.313 \cdot 0.1142 \cdot 0.25 \\ &\approx 0.00894 \end{aligned}$$

Since $f_S(x, y, z) > f_H(x, y, z)$, we predict **S (Silksong)** according to the maximum likelihood rule.

- (b) Now we need to apply the Bayes Decision Rule, which takes the prior probabilities into account. We must compute and compare

$$\begin{aligned} P(H \mid X = x, Y = y, Z = z) &\propto P(H) f_H(x) f_H(y) P(z=1|H) \\ &= 0.6 \cdot 0.397 \cdot 0.1123 \cdot 0.15 \\ &\approx 4.013 \times 10^{-3} \\ P(S \mid X = x, Y = y, Z = z) &\propto P(S) f_S(x) f_S(y) P(z=1|S) \\ &= 0.4 \cdot 0.313 \cdot 0.1142 \cdot 0.25 \\ &\approx 3.580 \times 10^{-3} \end{aligned}$$

Since the posterior probability for H is greater than the posterior probability for S , we predict **H (Hades 2)**.

2. (a) First we need to compute the marginal in Y. In the range $y \in [0, 1]$,

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f_{XY}(x, y) dx \\ &= \frac{6}{19} \int_0^2 x^2 + y^3 dx \\ &= \frac{6}{19} \left(\frac{1}{3} x^3 + x y^3 \right) \Big|_{x=0}^{x=2} \\ &= \frac{6}{19} \left(\frac{8}{3} + 2y^3 \right) \end{aligned}$$

So

$$f_Y(y) = \begin{cases} \frac{16}{19} + \frac{12}{19}y^3 & y \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

Then the conditional is the joint divided by the marginal:

$$f_{X|Y}(x|y) = \frac{\frac{6}{19}(x^2 + y^3)}{\frac{16}{19} + \frac{12}{19}y^3} = \frac{6x^2 + 6y^3}{16 + 12y^3}$$

For $0 \leq x \leq 2, 0 \leq y \leq 1$, 0 otherwise. Note that here, we think of y like a fixed number (the observed outcome of Y), and X as a random variable that depends on this number.

- (b) Now we compute the expected value $E(X | Y = y)$, which will be a function of y .

$$\begin{aligned} E(X | Y = y) &= \int_0^2 x \frac{6x^2 + 6y^3}{16 + 12y^3} dx \\ &= \frac{1}{16 + 12y^3} \left(\frac{6}{4} x^4 + 3x^2 y^3 \right) \Big|_{x=0}^{x=2} \\ &= \frac{24 + 12y^3}{16 + 12y^3} \end{aligned}$$

This is only valid for $0 \leq y \leq 1$, and is 0 otherwise.

- (c) Now we compute $E(X)$ using the law of iterated expectations. We need the marginal PDF of Y computed in part a.

$$\begin{aligned} E(X) &= E(E(X | Y)) \\ &= \int_0^1 \frac{24 + 12y^3}{16 + 12y^3} \cdot \left(\frac{16}{19} + \frac{12}{19}y^3 \right) dy \\ &= \int_0^1 \frac{12}{19} (2 + y^3) dy \\ &= \frac{12}{19} \left(2y + \frac{1}{4}y^4 \right) \Big|_{y=0}^{y=1} = \frac{12}{19} \cdot \frac{9}{4} \\ &= \frac{27}{19} \end{aligned}$$

Note that this agrees with the number we computed in HW6 problem 3b while figuring out the covariance.

3. We have $f_X(x) = 3e^{-3x}$ for $x \geq 0$, and $f_{Y|X}(y|x) = xe^{-xy}$ for $x, y \geq 0$. Thus the joint PDF is the marginal times the conditional:

$$f_{XY}(x, y) = f_{Y|X}(y|x)f_X(x) = \begin{cases} 3xe^{-x(y+3)} & x, y \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Now we integrate out X to get the marginal in Y :

$$\begin{aligned} f_Y(y) &= \int_{x=-\infty}^{\infty} f_{XY}(x, y) dx = \int_{x=0}^{\infty} 3xe^{-x(y+3)} dx \\ &= \left. \frac{-3e^{-x(y+3)}(x(y+3) + 1)}{(y+3)^2} \right]_{x=0}^{x=\infty} = \begin{cases} \frac{3}{(y+3)^2} & y \geq 0 \\ 0 & \text{otherwise} \end{cases} \end{aligned}$$

4. (a) The cost to Anthropic to service these N requests is:

$$\begin{aligned} E(Z \mid N) &= E(X_1 + X_2 + \cdots + X_N) \\ &= E(X_1) + E(X_2) + \cdots + E(X_N) \\ &= NE(X_i) \\ &= N \cdot \frac{1}{2}(1 + 4) \\ &= \frac{5}{2}N \end{aligned}$$

The second step follows from the fact that the X_i 's are all independent and identically distributed, and the third step follows from the information that this distribution is uniform on $[1, 4]$ dollars.

- (b) By the law of iterated expectations,

$$E(Z) = E(E(Z \mid N)) = E\left(\frac{5}{2}N\right) = \frac{5}{2}E(N)$$

What is $E(N)$? It is the expected number of requests in a span of 168 hours, where the interval between requests is exponentially distributed with mean 4 hours (i.e., $\lambda = \frac{1}{4}$). Thus the expected number of requests per week is $168 \cdot \frac{1}{4} = 42$ requests in 1 week. This means that N is a Poisson random variable with $\alpha = 42$, which means that $E(N) = \alpha = 42$. Then

$$E(Z) = \$\frac{5}{2} \cdot 42 = \$105$$

- (c) Since it costs Anthropic \$105 on average for their internal processing, they should charge \$205 per week to make a \$100 profit.